MACHINE LEARNING × ADVANCED PYTHON · FINAL PROJECT

Flag the students who need help, on day one

We predict a student's final Portuguese grade from what the school knows in September, and flag the ones likely to fail.

Shauryaman Singh · Chinmay Hajare
PGE 2 · 4 November 2026

01 THE PROBLEM

A missed student costs more than an extra meeting

USER	A school counsellor, in September
DECISION	Whom to offer support: tutoring, a meeting with the family
TARGET	Final grade G3, 0 to 20, pass at 10. Regression, then a yes/no flag

MISSED FAILING STUDENT

No help

and may fail the year: the costly mistake

UNNECESSARY FLAG

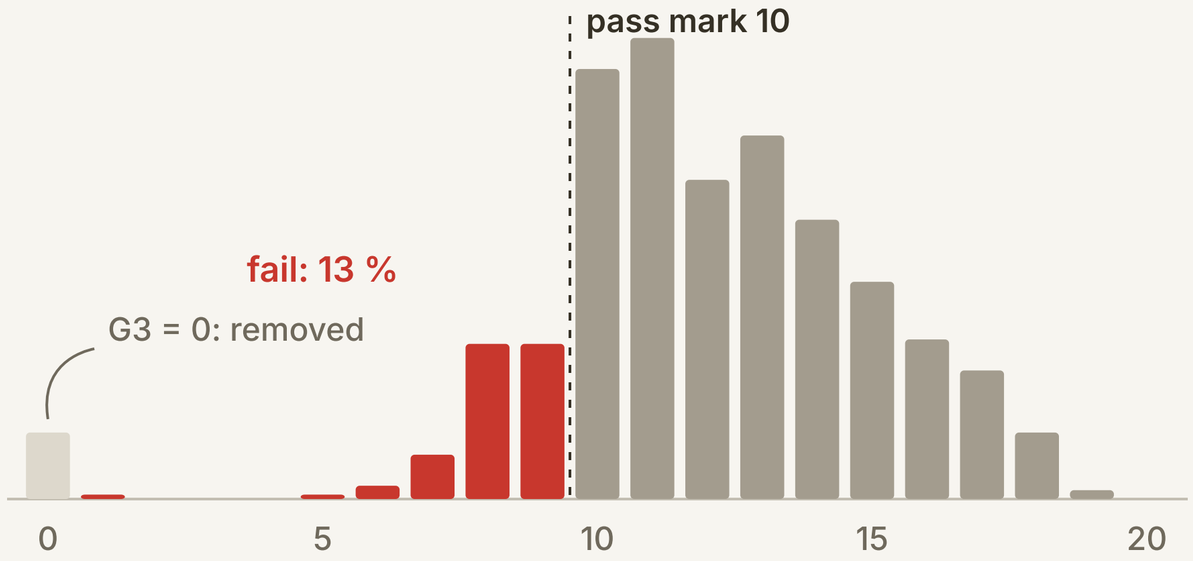
One meeting

a counsellor's time: the cheap mistake

> Our goal: flag **at least 75 %** of the students who will fail.

02 THE DATA

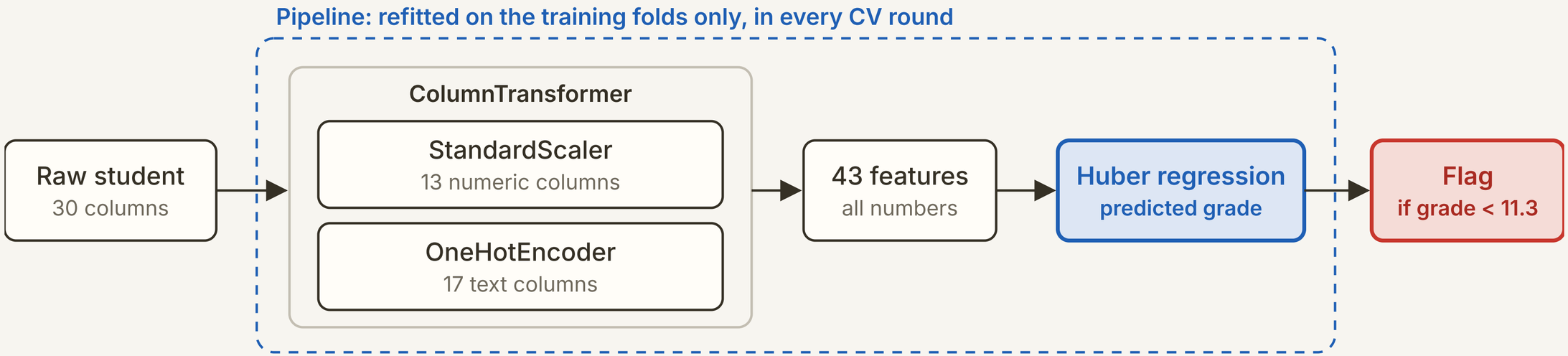
We removed what a counsellor can't know in September



Final grade G3 of the 649 students. UCI Student Performance (Cortez & Silva, 2008).

- **649 students**, 2 schools, 30 features, no missing values
- **15 with G3 = 0 removed:** 0 absences, they never sat the exam
- **G1, G2 dropped:** r = 0.92 with G3, but only known during the year (leakage)
- **Split once:** 507 train / 127 test, stratified on pass/fail, seed 42

One pipeline, so nothing leaks from the validation folds



`DatasetLoader`
data.py: load, clean, split

`Trainer`
models.py: CV, tuning, save, load

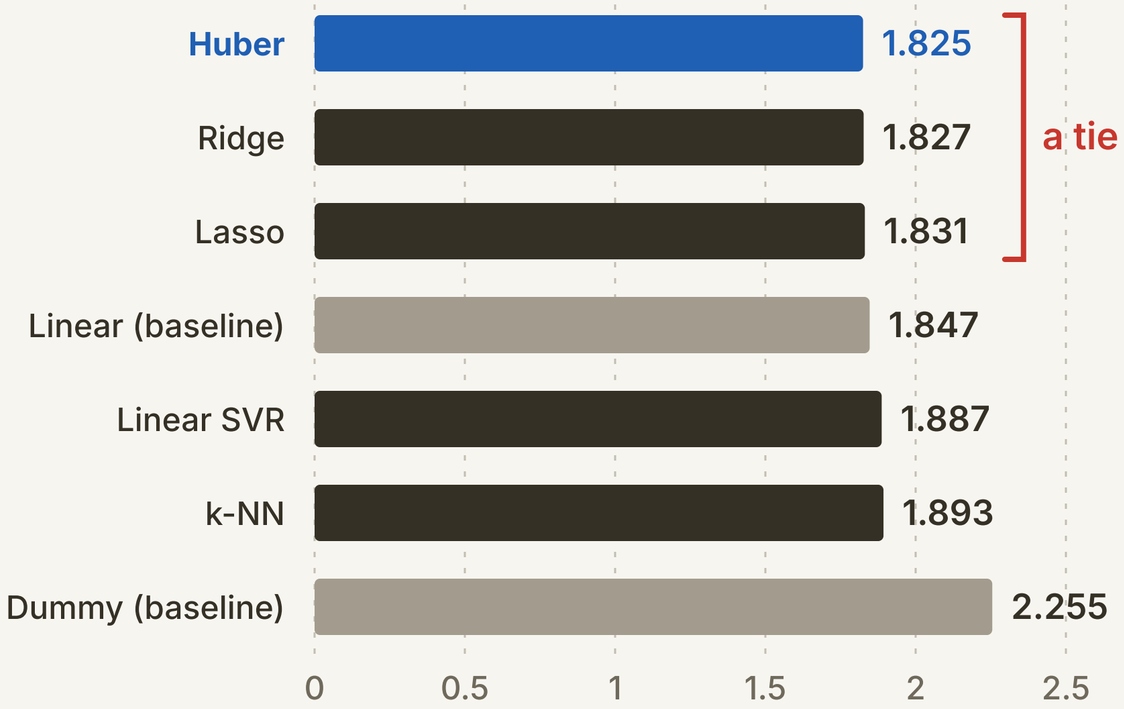
`cli.py`
train, evaluate, predict (argparse)

`config.yaml`
paths, seed, grids

`21 tests`
pytest, all passing

04 MODEL COMPARISON

Every model beats the baseline, and the linear ones tie



Mean absolute error in grade points, 5-fold CV on the 507 training students. Each model tuned with GridSearchCV.

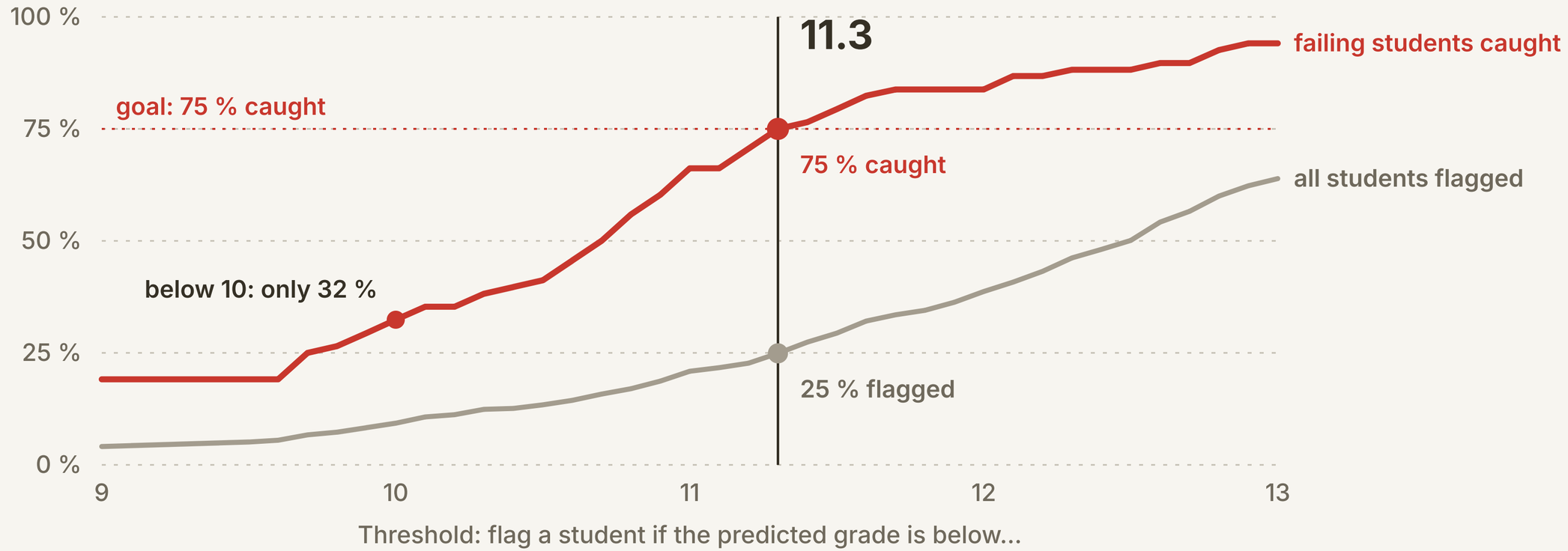
> The top three tie: gaps under 0.01, while the score moves ± 0.10 between folds.

WHY HUBER

- **Lowest CV MAE**, our rule fixed in advance
- **Robust:** large errors count linearly, not squared
- **Linear:** we can read its weights

05 FROM A GRADE TO A DECISION

Flagging below 11.3 catches 3 in 4 failing students



Out-of-fold predictions on the 507 training students: each one predicted by a model that never saw them.

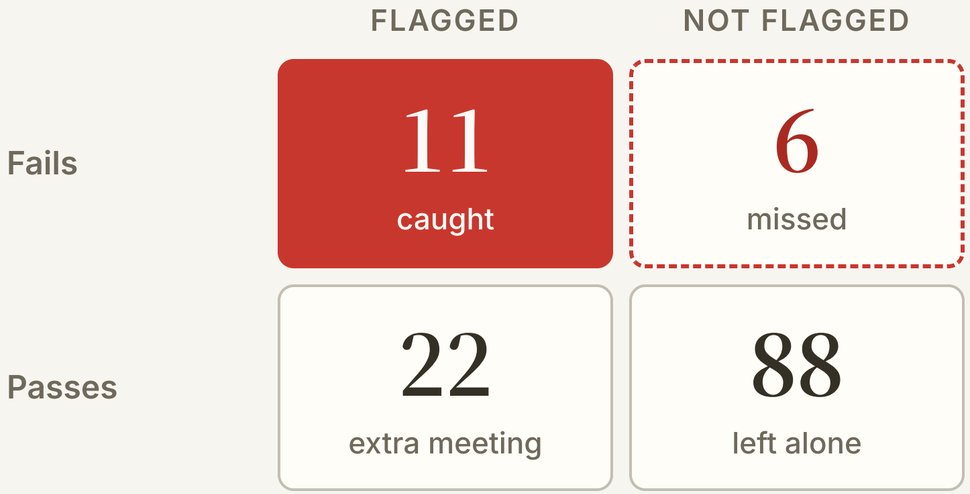
06 FINAL TEST, USED ONCE

On 127 unseen students: off by 1.7 points, and 11 of 17 failures caught

1.66

grade points, mean absolute error

RMSE 2.12 · in CV 1.83 · dummy 2.26



65 % recall

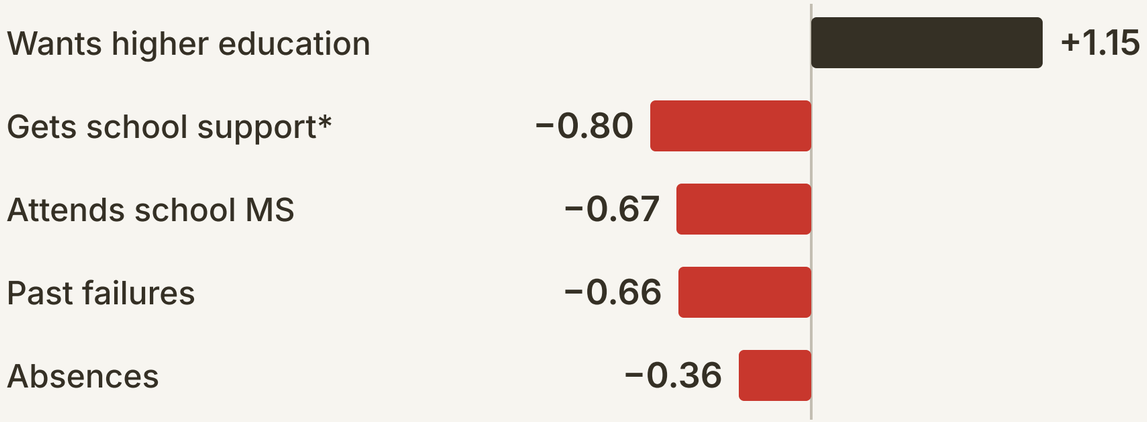
33 % precision

26 % flagged

07 WHAT IT MEANS FOR THE COUNSELLOR

A reason to talk to a student, never a verdict

STRONGEST SIGNALS (GRADE POINTS)



* a sign of struggle, not a cause

LIMITS

- About 1.7 points of error
- 634 students, 2 schools, 2005–06

RISKS

- 1 in 3 failing students missed
- Uses sex and family data: check fairness, GDPR

Live demo: `pytest` then `student-performance predict --input new_case.json`